What do programming languages do?

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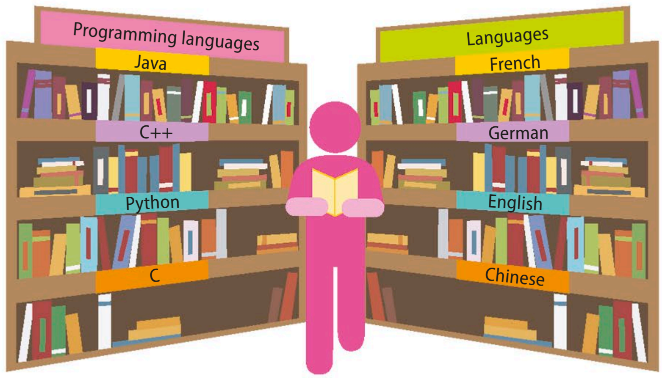
Programming languages were developed to help humans communicate with computers. The fundamental challenge is translating instructions humans can understand into ones computers can.

Programming languages

A programming language is a formalized set of words and symbols that allows people to give instructions to computers. Just like human languages, each programming language has its own vocabulary and grammar. Translating an algorithm written in English into a programming language enables a computer to understand and carry out the instructions.



▷ **Multilingual**
It's possible to translate text into many different human languages. Similarly, computer instructions can be written in many different programming languages.

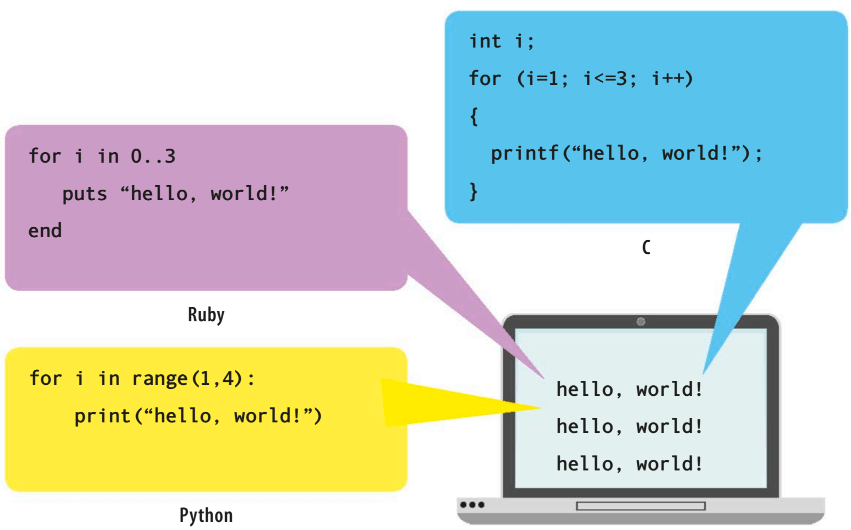
IN DEPTH

Translation

There is a variety of ways to translate programming languages into the binary code (sometimes called machine code) a computer understands. Some languages, like C and C++, use a compiler. This produces a new file containing machine code that can then be run. Scripting languages, like Python and JavaScript, use an interpreter, which translates and runs the code in a single process. Assembly languages use an assembler that, like a compiler, produces a file containing machine code.

Common features

All programming languages have certain underlying features. These are: making decisions, repeating instructions, and storing values in named containers. The words used for these features vary from language to language, but they're all doing essentially the same thing. Being familiar with these concepts in one language makes learning another language much easier.



▷ **Same outcome**
The three programs shown here look quite different, but they're all examples of a programming concept called “for loop” that is used to display a message three times.